Boundary-Aware Feature Fusion With Dual-Stream Attention for Remote Sensing Small Object Detection

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***Abstract*— Detecting small objects in remote sensing images poses significant challenges to the field of computer vision, primarily stemming from the complexity of backgrounds, limitations in pixel resolution, and information loss during the feature fusion process. While general object detection has significantly advanced in recent years, remote sensing small object detection remains an unsolved problem, with existing frameworks struggling to achieve high performance at small scales. In this article, we propose a novel framework called the boundary-aware feature fusion network (BAFNet), which significantly enhances the model's ability to represent and locate small objects precisely within complex remote sensing scenarios. First, a dual-stream attention fusion module captures complementary foreground and background cues through bidirectional context modeling. Jointly attending to objects and their surroundings enhances discriminative power for distinguishing small objects. Additionally, we incorporate a boundary-aware branch to better preserve crucial detailed information vital for small-scale objects. This auxiliary component supervises the fusion of contextual semantics and spatial information, aiding in retaining critical boundary details that are prone to loss during cross-layer feature fusion. We conducted experiments on the challenging AI-TOD, VisDrone, DIOR, and LEVIR-Ship datasets. The results demonstrate the superiority of our approach over other state-of-the-art (SOTA) object detection methods, particularly in terms of precisely identifying small objects within remote sensing images. The code is available at https://github.com/ooo1128/BAFNet.**

***Index Terms*— Boundary-aware, feature fusion, small object detection.**

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I. Introduction

SMALL object detection in remote sensing imagery has remained an important research focus, as it not only has extremely high academic value but also has broad application prospects in various domains, such as military reconnaissance, environmental monitoring, and urban planning. Its goal is to accurately identify specific objects in images by delineating their precise locations with bounding boxes and further determining their categories. In recent years, deep learning-based object detection models have been proposed and have made significant progress. The accurate detection of large- and medium-sized objects with high resolution and clear appearance has been achieved in many applications. However, small objects pose a significant challenge for detection because of their limited pixel ratio and sparse and blurred appearance. This characteristic makes it difficult to learn high-quality representations of small objects, resulting in existing object detection methods struggling to achieve the desired accuracy when detecting and locating small objects.

In view of the difficulties in small object detection, several studies have made important contributions to advancing the field. Some studies [1], [2], [3], [4], [5] have employed feature fusion methods to address the challenge of weak representation capability for small objects at a single feature level, enhancing the ability to capture and represent the characteristics of small objects, thereby improving detection accuracy. Researchers [6], [7], [8], [9], [10], [11] have focused on modeling the contextual relationships between small objects and their surrounding objects and environments. By capturing the inherent dependencies and interactions, these methods aim to enrich the representation of object features in images, leading to improved accuracy in small object detection. Furthermore, achieving balanced learning of all objects has been a key focus in improving detection performance. Some works [12], [13] have optimized label assignment strategies, enabling the model to allocate appropriate labels for each object more accurately, thus avoiding performance degradation caused by improper label assignment. As a crucial postprocessing step, nonmaximum suppression (NMS) has also been extensively studied to

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enhance the detection of overlapping objects. By improving the NMS algorithm, Luo et al. [14] effectively decreased false and missed detections, thereby enhancing both the precision and stability of the detection process.

Although the aforementioned methods have improved the performance of small object detection, their application in the field of remote sensing imagery still faces several limitations. The backgrounds of remote sensing images often encompass a wide variety of terrain and texture features, which can significantly interfere with the detection of small objects. Furthermore, small objects in high-resolution remote sensing images are typically composed of only a handful of pixels, resulting in blurred appearances and a lack of distinct internal features. Existing methods lack effective mechanisms to enhance the focus and recognition of target regions and often lose crucial detailed information during the feature extraction and fusion processes. This leads to unclear distinctions between objects and backgrounds, thereby affecting detection accuracy.

To address the aforementioned limitations, we propose a boundary-aware feature fusion network (BAFNet). Through a dual-stream attention mechanism and a boundary-aware supervision strategy, our method can effectively differentiate small objects from the background, enabling the precise detection of small objects in remote sensing images. Specifically, we utilize high-level semantic features to generate dual-stream attention maps, which are then combined with low-level spatial features to capture more discriminative foreground and background cues, effectively mitigating the issue of small objects being easily confused with the background. Furthermore, the boundary-aware supervision strategy guides the model to focus on boundary regions, effectively preventing the loss of important boundary information during feature fusion and subsequently enhancing the representation capabilities of the objects.

The key achievements of this study can be summarized as follows.

1) We propose a novel BAFNet object detection network that can address the challenge of small objects easily blending into complex backgrounds, resulting in poor detection performance. Experiments conducted on the AI-TOD, VisDrone, DIOR, and LEVIR-Ship datasets demonstrate that our proposed method achieves state-of-the-art (SOTA) performance.
2) We propose a dual-stream attention mechanism that leverages both foreground and background clues to augment the representation of object regions. The dual-stream attention module captures the mutually dependent features between the object and its encompassing context in a complementary learning manner, effectively capturing the salient features of small objects while suppressing interference from complex backgrounds.
3) We propose a boundary-aware supervision strategy that uses boundary information to guide the fusion of multiscale features, preserving fine-grained details and helping to better distinguish objects of interest from backgrounds, thereby more accurately localizing objects.

The subsequent sections of this article are structured as follows. In Section II, we outline the related work. Section III offers an overview and highlights of our proposed framework. In Section IV, we detail experimental evaluations on the AI-TOD, VisDrone, DIOR, and LEVIR-Ship datasets. Section V concludes the article on the basis of the research findings and provides prospects for future studies.

II. Related Work

A. Object Detection

Object detection in natural images has advanced significantly over the past decade and can be divided into two-stage detectors and one-stage detectors. Two-stage detectors [15], [16], [17] operate sequentially, initially identifying potential regions of interest (ROIs) through a region proposal stage. These ROIs then undergo classification and localization via convolutional neural networks. Early iterations relied on low-level image features, but subsequent advancements [18], [19] integrated region proposal networks (RPNs) for end-to-end training, enhancing performance. On the other hand, one-stage detectors [20], [21], [22], [23] bypass the RoI extraction stage and directly classify and regress bounding boxes via extracted convolutional features. This approach offers the advantage of faster detection speeds, making it well-suited for real-time applications. Recently, anchor-free detectors [24], [25] have gained popularity. These methods eliminate the need for manually defining anchor box sizes and corresponding matching protocols instead of tuning the detectors on the basis of the distribution of object scales within images.

Despite significant advancements in general object detection, there is still a performance gap between detecting small objects and detecting regular-sized objects because of the inherent characteristics of small objects. First, small objects are often confined to a handful of pixels, making extracting their informative features exceptionally challenging. This limited feature information can lead to the loss of crucial details, subsequently diminishing the detection accuracy. Second, small objects are prone to occlusion or concealment within the background, hindering the ability of detectors to identify and distinguish them accurately. Addressing these challenges is crucial for narrowing the performance gap in small object detection.

B. Small Object Detection in Remote Sensing Images

Feature pyramids have been widely employed in SOTA detectors to facilitate the detection of objects at various scales in remote sensing images. Lin et al. [1] introduced the feature pyramid network (FPN) in 2017, which enables detection algorithms to identify objects across various scales in images by constructing multiscale feature maps, significantly improving the accuracy of detecting small objects. Since then, many methods have attempted to further improve the FPN, such as PANet [2] and BiFPN [3]. Additionally, Cao et al. [26] proposed a multibranch and high-level semantic network (MHN) to address the issue of inconsistent semantic levels among different layers of features, significantly increasing

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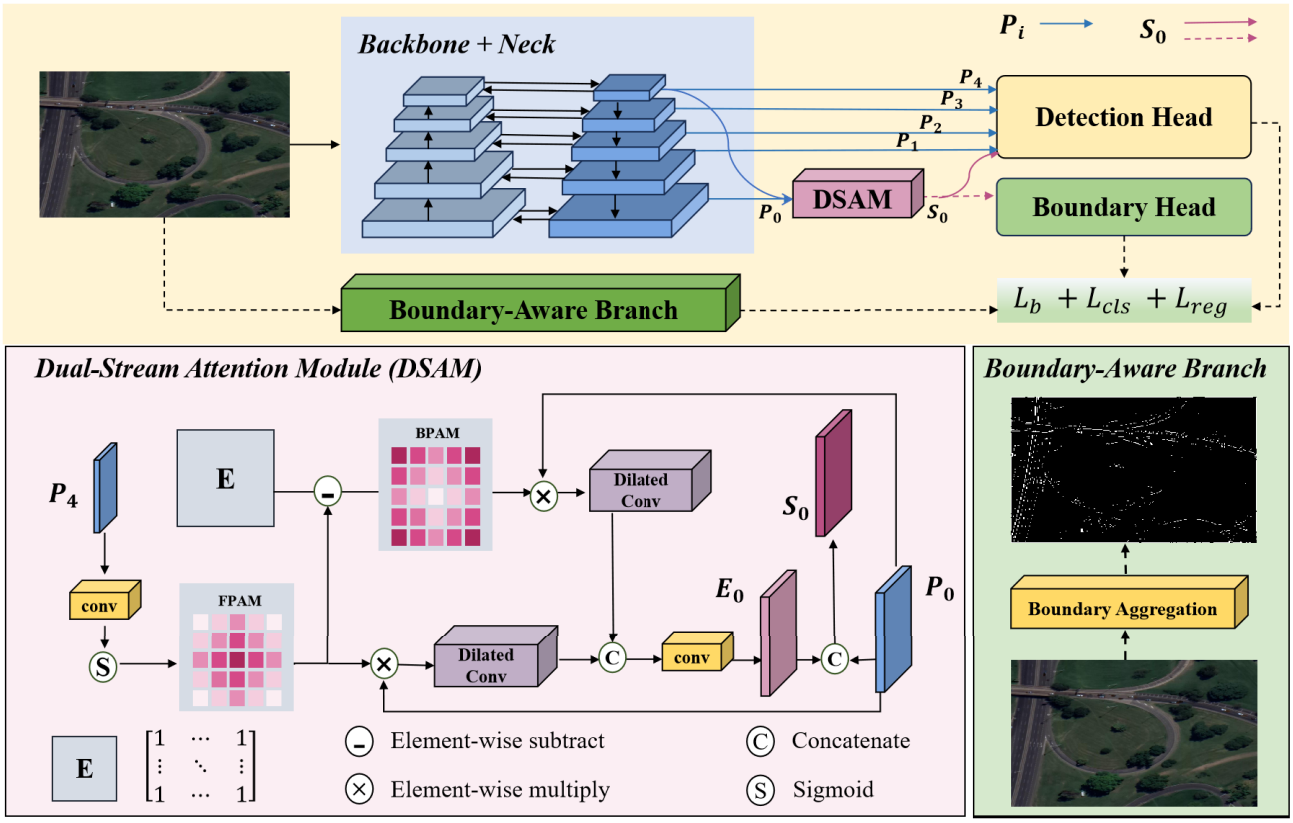

Fig. 1. Framework of our proposed method. In BAFNet, the DSAM enhances small object features by generating foreground and background attention maps from high-level feature pyramid outputs. These attention maps are fused with spatial features to yield enhanced representations. Additionally, a boundary-aware branch assists fusion via boundary loss, strengthening the preservation of object boundaries during small object detection. The dashed part of the figure is involved only in the training process.

the performance of small object detection. Sun et al. [27] proposed a high-resolution network (HRNet), which enhances feature representations through repeated cross-stage partial connections for multiscale fusion. Liu et al. [28] proposed the adaptive structure feature fusion (ASFF), which adaptively predicts the weights of features at different stages in the refusion process through an adaptive mechanism. Wang et al. [29] introduced pyramid pooling to improve the efficiency of feature fusion between adjacent layers. Ghiasi et al. [30] explored the optimal combination of feature fusion across layers via AutoML. Qiao et al. [31] proposed a novel object detection framework DetectoRS, which significantly improves detection performance by introducing a recursive feature pyramid (RFP) at the macrolevel and a switchable atrous convolution at the microlevel. Wu et al. [32] introduced a feature-and-spatial aligned network (FSANet), which incorporates a feature-aware alignment module to enhance multiscale features and a spatial-aware guidance head to iteratively refine object localization. This addresses the issues of feature scarcity and subpar localization in detecting minute objects within remote sensing imagery. Zhou et al. [33] proposed an adaptive online training sample mining strategy that incorporates task-level attention to guide the model in selectively choosing suitable features for classification or regression tasks, thereby improving detection performance for small objects.

The aforementioned methods have further improved the effectiveness of feature fusion in various ways. However, these methods generally overlook the importance of boundary detail information. Specifically, for small objects, directly fusing high-level semantic context information with low-level spatial information may lead to issues such as boundary-blurring or over-erosion. This results in the loss of detailed features, ultimately affecting detection performance.

III. Methodology

In remote sensing images, small objects are prone to blending with the background owing to their low pixel ratio, making it difficult to extract discriminative features. To improve detection accuracy, we propose an effective detection framework, BAFNet, which is specifically designed for small object detection. The core of this framework lies in the use of a dual-stream attention mechanism to capture foreground and background cues for extracting distinctive object features and the introduction of image boundary information as auxiliary supervision to increase performance. The framework of our method is shown in Fig. 1. First, the input image is fed into the backbone and neck to extract multiscale feature maps. Then, the dual-stream attention module fuses the highest-level semantic features with low-level spatial features to distinguish the foreground from the background, obtaining more discriminative features. All feature maps of different scales are subsequently input into the detection head, which predicts the final category and bounding box of the objects. During training, the enhanced low-level features are fed into the boundary head to predict the boundary. Moreover, the boundary-aware branch extracts boundary information from

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the input image as the ground truth, which is used together with the predicted boundary in network training to further improve the performance of small object detection.

A. Backbone and Neck

The input image I first passes through the backbone and neck, outputting multiscale feature maps of different resolutions, labeled $P_i (i = 0, 1, 2, 3, 4)$. In our method, we adopt pretrained ResNet50 [34] as the backbone network for extracting multiscale features of the image, whereas the RFP [31] serves as the neck network responsible for the feature fusion process.

B. Dual-Stream Attention Module

The dual-stream attention module aims to distinguish the foreground from background regions, obtaining more discriminative features, as shown in Fig. 1. Given that semantic information enables a better understanding of the context and relationships between different objects and regions in an image, it is crucial for accurately differentiating objects from the background. Consequently, we utilize the richest semantic top-level feature P_4 to generate the foreground priority attention map (FPAM), which emphasizes the foreground of the object. Following sigmoid activation, we subtract the FPAM from an all-one matrix E to generate the background priority attention map (BPAM):

$$\begin{cases} \mathbf{A}^F = \sigma(U(\text{conv}(\mathbf{P}_4))) \\ \mathbf{A}^B = \mathbf{E} - \mathbf{A}^F \end{cases} \tag{1}$$

where $\mathbf{A}^F$ represents FPAM, $\mathbf{A}^B$ represents BPAM, $U(\cdot)$ represents upsampling, conv$(\cdot)$ is the convolution operation, where a 1×1 convolution kernel is used, and $\sigma(\cdot)$ is a sigmoid function.

We then weight fuse FPAM and BPAM with low-level feature P_0 to further incorporate fine-grained details, enhancing the representation of small objects in the feature map and making them more discriminative. Notably, to enhance the perception ability for small objects, we introduce multiscale convolution to capture the global object region

$$\mathbf{E_0} = \mathrm{L}\left(\left[\sqcup_{i=1}^{4} \mathcal{D}_i\left(\mathbf{P}_0 \odot \mathbf{A}^F\right), \sqcup_{i=1}^{4} \mathcal{D}_i\left(\mathbf{P}_0 \odot \mathbf{A}^B\right)\right]\right) \tag{2}$$

$$\mathbf{S}_0 = \left[\mathbf{E_0}, \mathbf{P}_0\right] \tag{3}$$

where $\odot$ denotes elementwise multiplication. $\sqcup$ denotes the concatenation operation. $\mathcal{D}_1$ is composed of 1×1 kernels with 256 channels, followed by a ReLU layer. $\{\mathcal{D}_i\}_{i=2}^{4}$ is a set of dilated convolutions with rates of 3, 5, and 7, as shown in Fig. 2. They all have 3×3 kernels with 256 channels followed by a ReLU layer. The two branches do not share parameters. $[\cdot, \cdot]$ denotes concatenation. L represents the output layer, where the concatenated features are passed through a 3×3 kernel and ReLU layer to generate enhanced features S_0.

C. Boundary-Aware Auxiliary Branch

The purpose of the boundary-aware auxiliary branch is to use boundary information to supervise the fusion of spatial details and semantic context, preventing the loss of fine-grained details of small objects during cross-scale feature fusion. We generate binary boundary ground truths via the boundary aggregation module, as shown in Fig. 1. Given the broad coverage, diverse objects, and varying scales in remote sensing images, capturing multiscale boundary features is particularly crucial. Therefore, we employ the Laplacian operator to extract boundary features at different strides, constructing a Laplacian pyramid to obtain multiscale boundary information, as illustrated in Fig. 3

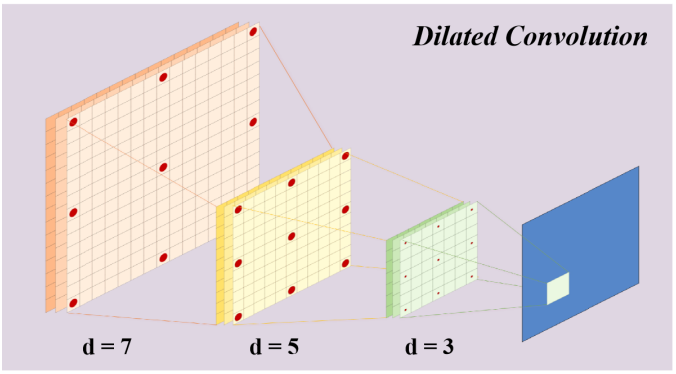

Fig. 2. Dilated convolution. "d" denotes the rate of dilation.

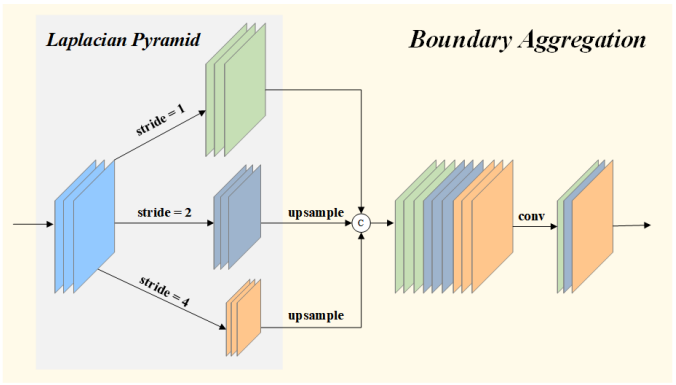

Fig. 3. Boundary aggregation.

$$\begin{aligned} r_1 &= L_1(g_d) \\ r_2 &= L_2(g_d) \\ r_4 &= L_4(g_d) \end{aligned} \tag{4}$$

where g_d is the input image after grayscale conversion. L_i represents Laplacian convolution with a downsampling stride of i, which consists of a 2-D convolution and a trainable 1×1 convolution. The multiscale boundary representations $r_i (i = 1, 2, 4)$ obtained from g_d are concatenated as follows:

$$g_p = [r_1, U(r_2), U(r_4)] \tag{5}$$

where U represents the upsampling operation, the multiscale boundary representations $r_i (i = 1, 2, 4)$ are adjusted to the same dimension via bilinear interpolation and concatenated into a boundary feature map g_p.

Afterward, we resize the boundary feature maps to their original dimensions and adaptively adjust their weights through integration with a learnable 1×1 convolutional

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layer

$$f_b = \mathrm{conv}(g_p). \tag{6}$$

Finally, we convert the predicted boundary maps f_b to the final binary boundary ground truth via a threshold of 0.1.

Moreover, we introduce an additional boundary detection head to predict boundaries. Then, the obtained boundary ground truth is used as guidance to instruct the network in learning to preserve the boundary details of the objects.

D. Loss Function

We optimize the detection branch and boundary branch concurrently. The total loss function is defined as follows:

$$\mathcal{L} = \alpha \mathcal{L}_b + \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{reg}} \tag{7}$$

where $\mathcal{L}_{\mathrm{cls}}$ and $\mathcal{L}_{\mathrm{reg}}$ are the original classification loss and regression loss of the detection head, respectively. $\mathcal{L}_b$ is the loss of the boundary branch, and the parameter α is used to scale its weight.

1) Boundary Loss: Since rough boundaries can better highlight the boundary regions of small objects and enhance their feature representation, we employ the weighted binary cross-entropy loss function for optimizing boundary learning. This loss function assigns different weights to pixels at different positions on the feature map, with higher weights assigned to pixels at the boundary positions and lower weights assigned to pixels farther from the boundary. Therefore, the boundary loss L_b can be expressed as

$$L_b = -\frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\left(1+\gamma\alpha_{ij}\right)\sum_{l=0}^{1} 1\left(g_{ij}=l\right)\log \Pr\left(p_{ij}=l \mid \Psi\right)}{\sum_{i=1}^{H}\sum_{j=1}^{W}\gamma\alpha_{ij}} \tag{8}$$

where $\mathbf{1}(\cdot)$ denotes an indicator function, while γ serves as a tunable hyperparameter. The symbol l takes on values of 0 or 1, signifying two distinct label categories. p_{ij} and g_{ij} represent the predicted and actual values, respectively, for the pixel located at (i, j) in the image. The collective parameters of the model are denoted by Ψ, and $\Pr(p_{ij} = l \mid \Psi)$ signifies the predicted probability of the pixel belonging to label l.

The complete process of BAFNet is shown in Algorithm 1. We use ResNet and RFP for multiscale feature extraction and fusion, obtaining features $V_i (i = 0, 1, 2, 3, 4)$ at different scales. To enhance the foreground representation of small objects, we further fuse the highest-level feature f_4 with rich semantic information and the low-level feature f_0 containing more spatial details through the dual-stream attention module (DSAM) module, obtaining the enhanced feature S_0. We extract the boundary features X_b of the input image I via a Laplacian pyramid, input S_0 into the boundary head to obtain the predicted boundaries bd, and compute the loss between them to supervise the aforementioned fusion process. Finally, S_0 and $P_i (i = 1, 2, 3, 4)$ are fed into the detection head to obtain the object detection results in (cls, box).

Algorithm 1 Algorithm for BAFNet

Input: Images and their corresponding annotations.
Output: Position and label of the object in the image.
Let $V_i (i = 0, 1, 2, 3, 4)$ denote the multiscale features extracted by ResNet, $P_i (i = 0, 1, 2, 3, 4)$ represent the multiscale features fused by RFP, I denote the input, $DHead$ denote the object detection head, and $Bhead$ denote the boundary detection head.
for *epoch*=1 **to** E **do**
  $V_i = ResNet(I)$;
  $P_i = RFP(V_i)$;
  $S_0 = DSAM(P_0, P_4)$;
  $X_l = (S_0, P_1, P_2, P_3, P_4)$;
  $X_b = Laplacian(I)$;
  $bd = BHead(X_l)$;
  $cls, box = DHead(X_l)$.
end

IV. Experiment

A. Experimental Setup

1) Datasets: To demonstrate the effectiveness of our approach, we conduct experiments on four datasets: AI-TOD [43], VisDrone2019 [63], DIOR [64], and LEVIR-Ship [61].

AI-TOD is an optical remote sensing dataset specifically designed for small object detection. It consists of a total of 28 036 images with a size of 800 × 800 pixels, split into training (11 214), validation (2804), and testing (14 018) sets. AI-TOD encompasses eight common object categories, namely, airplanes (AIs), bridges (BRs), storage tanks (STs), ships (SHs), swimming pools (SPs), vehicles (VEs), people (PEs), and windmills (WMs). The AI-TOD dataset features an average object size of 12.8 pixels, which is markedly smaller than benchmarks in natural scenes such as MS COCO [65] (99.5 pixels) and PASCAL VOC [66] (156.6 pixels). This small object size poses great challenges for object detection algorithms when they are applied to optical remote sensing imagery.

The VisDrone2019 dataset focuses on scenes captured by unmanned aerial vehicles (UAVs), encompassing a total of 8599 images. The dataset is divided into training, validation, and test sets, containing 6471, 548, and 1580 images, respectively. It provides annotations for 11 different classes, namely, pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, motor, and others (representing nonvalid object regions). One of the unique strengths of the VisDrone dataset lies in its diversity in terms of locations, environments, objects, and densities.

DIOR is a dataset specifically designed for object detection in optical remote sensing images. It contains 23 463 images and 192 472 object instances, covering 20 object categories such as airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf course (GC), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA),

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storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM). This dataset is large in scale in terms of the number of object categories, the number of object instances, and the total number of images. It features a wide range of object sizes, diverse image variations, high interclass similarity, and intraclass diversity, providing comprehensive challenges for object detection in remote sensing images.

The LEVIR-Ship dataset is a large-scale dataset specifically tailored for the detection of tiny ships in remote sensing images. It contains 3896 images, each with a resolution of 512 × 512 pixels, encompassing over 3000 instances of tiny ships. The pixel size of the ship objects is mostly less than 20 × 20 and is concentrated around 10 × 10, making it particularly suitable for the task of small object detection in remote sensing scenarios.

2) Implementation Details: We implemented our method on the PyTorch framework [67] and MMDetection [68]. We followed the default data augmentation settings in MMdetection. We chose DetectoRS [31] as the detector of our proposed model and adopted the RFLA [13] label assignment strategy. Optimization was performed via the SGD optimizer, which was configured with a momentum of 0.9 and a weight decay factor of 0.0001. We set the batch size to 2 and set the learning rate to 0.005. Additionally, we set the RPN proposal number to 3000. For the experiments on the AI-TOD dataset, we train for 36 epochs, which decay at the 24th and 33th epochs. We trained for 12 epochs on the VisDrone, DIOR, and LEVIR-Ship datasets, with learning rate decay applied at the 8th and 11th epochs. During inference, background bounding boxes were filtered out using a confidence threshold of 0.05, and NMS was applied with an IoU threshold of 0.5, limiting the maximum number of bounding boxes to 3000. In the loss function 7, we set the balance factor as $\alpha = 0.5$.

3) Evaluation Metrics: To evaluate the proposed detection algorithm, we employed the same evaluation metrics as outlined in MS COCO [65] and used the average precision (AP) metric to assess the algorithm's performance. The AP metric is formally defined as

$$\mathrm{AP} = \int_0^1 P(r)dr \tag{9}$$

where r signifies the recall rate and where $P(r)$ represents the precision at a given recall level. The recall rate and precision are derived via the following equations:

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

where true positive (TP) refers to the number of correctly matched samples under different IoU thresholds, false positive (FP) represents the number of incorrectly matched samples, and false negative (FN) denotes the number of missed samples.

We also calculate the detection scores of objects under different thresholds. Specifically, AP_{50} represents the AP calculated when the IoU threshold for defining the TP is 0.5. AP_{75} represents the AP calculated when the IoU threshold is 0.75. The AP is defined as the mean of AP_{50} to AP_{95}, where the IoU thresholds vary from 0.5 to 0.95 with a step of 0.05 when the TP is computed for computing the metric. In addition, we calculate the detection scores for objects of different volumes to determine the performance of our method in detecting small objects. $\mathrm{AP}_{\mathrm{vt}}$ refers to the AP calculated for objects with a volume smaller than 8^2 pixels. AP_t refers to the AP for objects with a volume between 8^2 and 16^2 pixels. AP_s refers to the AP for objects with a volume between 16^2 and 32^2 pixels. AP_m refers to the AP for objects with a volume greater than 32^2 pixels.

B. Comparison With SOTA Methods

To assess the effectiveness of the proposed BAFNet, we conducted a comparative evaluation against leading object detection models on the AI-TOD, VisDrone, DIOR, and LEVIR-Ship datasets, including several remote sensing algorithms renowned for their SOTA capability in detecting small objects.

1) Experiment on AI-TOD: Table I shows the experimental results of BAFNet compared with those of other SOTA detectors on the AI-TOD dataset. Specifically, these detectors include some of the most advanced methods for remote sensing images [32], [33], [44], [45], [46], as well as other leading detectors designed for general CV tasks. Our method achieves the best detection performance, with an AP of 30.5. Notably, our method demonstrates a significant advantage in detecting very tiny objects, achieving an $\mathrm{AP}_{\mathrm{vt}}$ of 16.6%, which is a 3.8% increase over the SOTA methods. This indicates that our method excels in addressing the challenge of detecting small objects in remote sensing scenarios.

As evident in Table I, the performance of the general detectors is below 7 $\mathrm{AP}_{\mathrm{vt}}$, with some even achieving 0 $\mathrm{AP}_{\mathrm{vt}}$. We attribute the reasons to the following two points.

1) Most models are trained on a commonly used COCO dataset where the number of small objects is limited, and the models are targeted for general object detection. Therefore, their performance is not satisfactory for AI-TOD.
2) During the process of feature extraction and fusion, it is highly prone to losing fine-grained feature information of very small objects, leading to poor performance in the $\mathrm{AP}_{\mathrm{vt}}$ metric. In comparison, our method can achieve higher AP values, especially $\mathrm{AP}_{\mathrm{vt}}$, while still maintaining good performance in detecting medium- and large-sized objects.

Fig. 4 shows a series of qualitative detection results across diverse scenes, underscoring the efficacy of our method in detecting small objects, including extremely tiny objects. Within urban environments, our method can accurately identify small objects such as vehicles and tanks, even when they are densely occluded or in cluttered backgrounds. In natural landscapes, our method demonstrates exemplary performance in detecting tiny objects in complex terrains. Aerial images show that our method is capable of identifying aircraft hidden

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TABLE I

RESULTS OF OUR PROPOSED BAFNET COMPARED WITH THOSE OF SOTA APPROACHES ON THE AI-TOD DATASET. RED AND BLUE: TOP TWO PERFORMANCES

Method	Backbone	AI	BR	ST	SH	SP	VE	PE	WM	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
TridentNet [35]	ResNet-50	7.6	0.0	10.8	9.6	2.5	7.0	1.9	0.0	4.9	13.1	2.7	0.0	2.8	10.8	19.2
RetinaNet [20]	ResNet-50	0.4	0.0	2.4	10.0	0.0	5.8	1.0	0.0	2.4	8.4	0.7	0.8	2.7	4.0	8.8
AutoAssign [36]	ResNet-50	14.7	10.7	25.5	26.4	4.1	16.5	6.0	3.0	13.4	36.2	6.9	3.5	13.5	17.0	24.9
Mask R-CNN [16]	ResNet-50	20.0	1.9	17.4	18.6	8.0	10.9	3.8	0.0	10.1	22.7	7.6	0.0	5.8	22.1	32.6
CenterNet [24]	ResNet-18	12.8	2.0	15.3	15.9	1.7	11.7	4.6	1.5	8.2	24.3	3.4	1.3	6.9	12.9	22.3
Dynamic R-CNN [37]	ResNet-50	10.4	15.5	30.7	38.7	7.0	19.5	9.4	3.9	16.9	39.4	11.6	6.3	18.2	21.2	25.6
TOOD [38]	ResNet-50	22.2	5.3	28.9	27.9	9.1	18.6	7.0	0.5	14.9	34.7	10.6	3.3	12.8	21.7	33.1
ATSS [39]	ResNet-50	0.5	9.8	16.8	26.4	0.6	12.0	4.0	1.2	8.9	22.5	5.3	2.1	10.1	11.4	15.0
Cascade R-CNN [17]	ResNet-50	21.0	6.7	20.3	21.0	7.7	12.8	4.8	0.0	11.8	25.6	9.5	0.1	7.9	23.0	34.8
Faster R-CNN [18]	ResNet-50	19.6	1.4	17.3	17.4	7.9	10.4	3.6	0.0	9.7	22.1	6.8	0.0	5.2	21.7	32.5
DetectoRS [31]	ResNet-50	25.3	8.2	22.9	25.4	13.2	15.2	5.9	0.1	14.5	32.2	10.9	0.1	10.6	27.7	37.7
FCOS [25]	ResNet-50	17.2	1.6	21.2	19.8	0.8	13.3	4.9	0.0	9.8	24.1	6.2	1.2	7.8	16.1	27.2
PANet [2]	ResNet-50	19.8	1.4	17.0	17.5	7.7	10.4	3.7	0.0	9.7	22.2	6.8	0.0	5.4	21.6	32.5
RepPoints [40]	ResNet-50	0.0	0.0	12.7	16.1	0.0	10.6	2.4	0.0	5.2	15.2	2.2	1.3	5.4	7.5	13.7
FoveaBox [41]	ResNet-50	14.7	0.0	19.2	17.7	0.0	12.4	4.2	0.0	8.5	20.6	5.7	0.9	6.1	14.1	27.2
SSD-512 [21]	VGG-16	10.9	2.1	10.7	14.3	1.8	9.2	2.2	0.9	6.5	21.0	2.4	1.0	5.3	11.9	21.2
YOLOv3 [42]	DarkNet-53	19.1	9.7	21.7	20.9	4.7	14.5	5.2	2.7	12.3	35.3	5.6	3.2	12.0	18.5	24.2
M-CenterNet [43]	DLA-34	18.6	10.6	27.6	22.3	7.5	18.6	9.2	2.0	14.5	40.7	6.4	6.1	15.0	19.4	20.4
KLDNet [33]	ResNet-50	13.6	18.7	35.7	42.3	5.2	24.9	9.3	6.7	19.6	-	-	8.4	20.6	22.7	26.4
FSANet [32]	Swin-T	30.9	15.1	35.0	40.3	19.8	24.9	8.9	5.6	22.6	52.8	15.6	7.4	21.6	29.1	38.5
MENet [44]	Swin-T	27.5	16.4	37.4	42.1	18.8	24.9	10.2	8.2	23.2	56.2	15.0	9.7	23.9	25.3	34.4
DNTR [45]	ResNet-50	-	-	-	-	-	-	-	-	26.2	56.7	20.2	12.8	26.4	31.0	37.0
CAF^{2}ENet-N [46]	DarkNet-53	29.0	12.9	35.2	34.6	8.3	27.2	10.4	3.7	20.2	48.1	13.3	6.0	19.9	26.5	29.7
CAF^{2}ENet-T [46]	DarkNet-53	33.2	16.8	40.4	42.8	21.3	33.3	14.1	4.5	25.8	57.3	19.9	10.8	26.0	32.6	36.0
CAF^{2}ENet-S [46]	DarkNet-53	38.5	18.2	41.5	45.8	22.9	34.5	15.8	7.7	28.1	61.4	21.8	12.4	27.9	34.9	39.2
CAF^{2}ENet-M [46]	DarkNet-53	39.7	22.8	43.7	50.6	24.6	34.9	17.1	8.9	30.2	63.7	25.0	12.8	30.3	36.7	41.9
Ours	ResNet-50	34.5	22.7	39.4	68.1	29.3	28.1	14.9	6.9	30.5	59.8	26.6	16.6	31.3	35.1	40.5

Fig. 4. Qualitative results of our method on the AI-TOD datasets.

in clouds, despite their small proportions. Across these challenging scenarios, the proposed method exhibits robust small object detection capabilities, demonstrating its effectiveness in environments with complex backgrounds.

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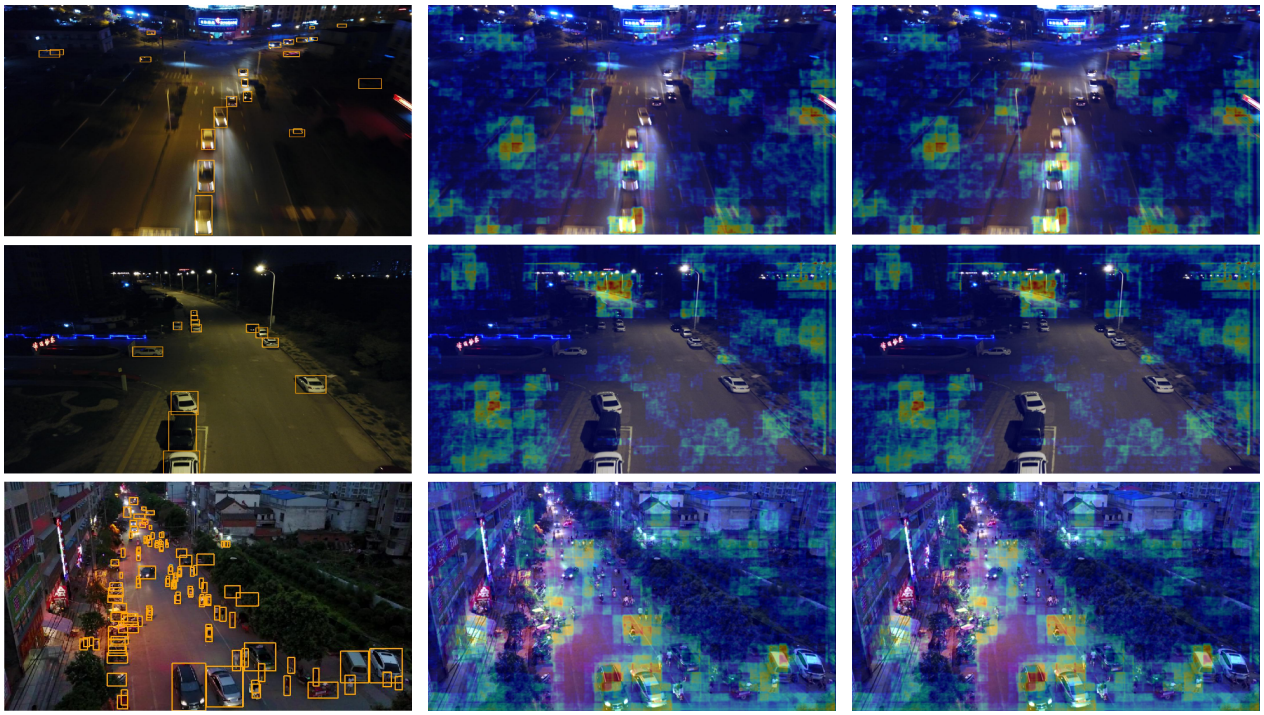

(a) **Image** (b) **Baseline** (c) **Ours**

Fig. 5. Qualitative comparison plot of the VisDrone dataset. (a) Original input image. (b) Heatmap of the bottom features of the baseline. (c) Heatmap of the bottom features of BAFNet.

TABLE II

RESULTS OF OUR PROPOSED BAFNET COMPARED WITH THOSE OF SOTA APPROACHES ON THE VISDRONE DATASET. RED AND BLUE: TOP TWO PERFORMANCES

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
RetinaNet [20]	17.6	29.6	18.1	8.1	29.4	36.8
AutoAssign [36]	23.2	43.5	21.8	15.2	33.3	40.9
GFL V1 [47]	23.4	38.2	24.2	14.3	34.7	40.1
CenterNet [24]	21.4	36.1	21.8	12.6	31.7	38.3
YOLOF [48]	15.1	26.3	15.4	6.1	25.2	32.4
Cascade R-CNN [17]	25.6	43.1	26.2	16.4	37.4	41.4
Faster R-CNN [18]	24.5	42.5	24.6	16.1	35.9	36.5
PANet [2]	23.0	38.1	24.2	13.9	34.6	39.0
CMDNet [49]	29.2	49.5	29.8	16.1	35.9	36.5
QueryDet [50]	28.3	48.1	28.7	-	-	-
DINO [51]	26.8	44.2	28.9	17.5	37.3	41.3
RT-DETR [52]	15.9	36.5	10.7	13.8	28.4	23.1
EF-DETR [53]	13.2	24.2	12.3	8.1	18.4	35.7
Ours	30.8	52.6	30.9	22.4	41.0	43.2

2) Experiment on VisDrone: Table II provides a detailed performance comparison of our method and several SOTA approaches on the VisDrone validation set. For two-stage detectors, we compared BAFNet against Cascade R-CNN [17], Faster R-CNN [18], PANet [2], CMDNet [49], and QueryDet [50]. Additionally, we compared our approach with one-stage detectors, including RetinaNet [20], AutoAssign [36], GFL V1 [47], CenterNet [24], and YOLOF [48]. Furthermore, a comprehensive evaluation of performance was conducted against transformer-based object detectors DINO [51], RT-DETR [52], and EF-DETR [53]. The results indicate that BAFNet outperforms the other methods in terms of all metrics. Notably, BAFNet achieves an AP value of 30.8%, surpassing that of CMDNet, the top performer, by 1.6% points, demonstrating its superiority in terms of detection accuracy. Particularly in detecting small objects, BAFNet achieved an AP_s value of 22.4%, outperforming DINO, the next best performer, by 4.9 percentage points, showcasing BAFNet's strength in handling tasks related to small object detection.

Additionally, visual comparisons of different methods are shown in Figs. 5 and 6. In Fig. 5, we visualize the heatmaps of the baseline and our method for better analysis and understanding of the advantages of our method. Specifically, compared with the baseline, our method more effectively focuses attention on foreground objects and significantly enhances the response intensity of small objects. Further observation of Fig. 6 clearly demonstrates the exceptional accuracy of our BAFNet in detecting small objects, effectively addressing the challenges posed by scale variations. The regions highlighted by yellow dashed lines particularly emphasize BAFNet's powerful ability in small object detection, accurately identifying and locating small objects that are often missed or difficult to detect by other methods. Moreover, the regions marked by red dashed lines also validate the superior performance of BAFNet in detecting occluded objects. In these regions, the objects are often partially occluded by other objects or complex backgrounds, which increases the difficulty of detection. Nevertheless, our BAFNet successfully captures occluded objects by integrating information through the proposed boundary-aware attention fusion strategy. These phenomena not only prove the effectiveness of BAFNet in performing small object detection tasks but also demonstrate its ability to handle complex scenes and challenging objects.

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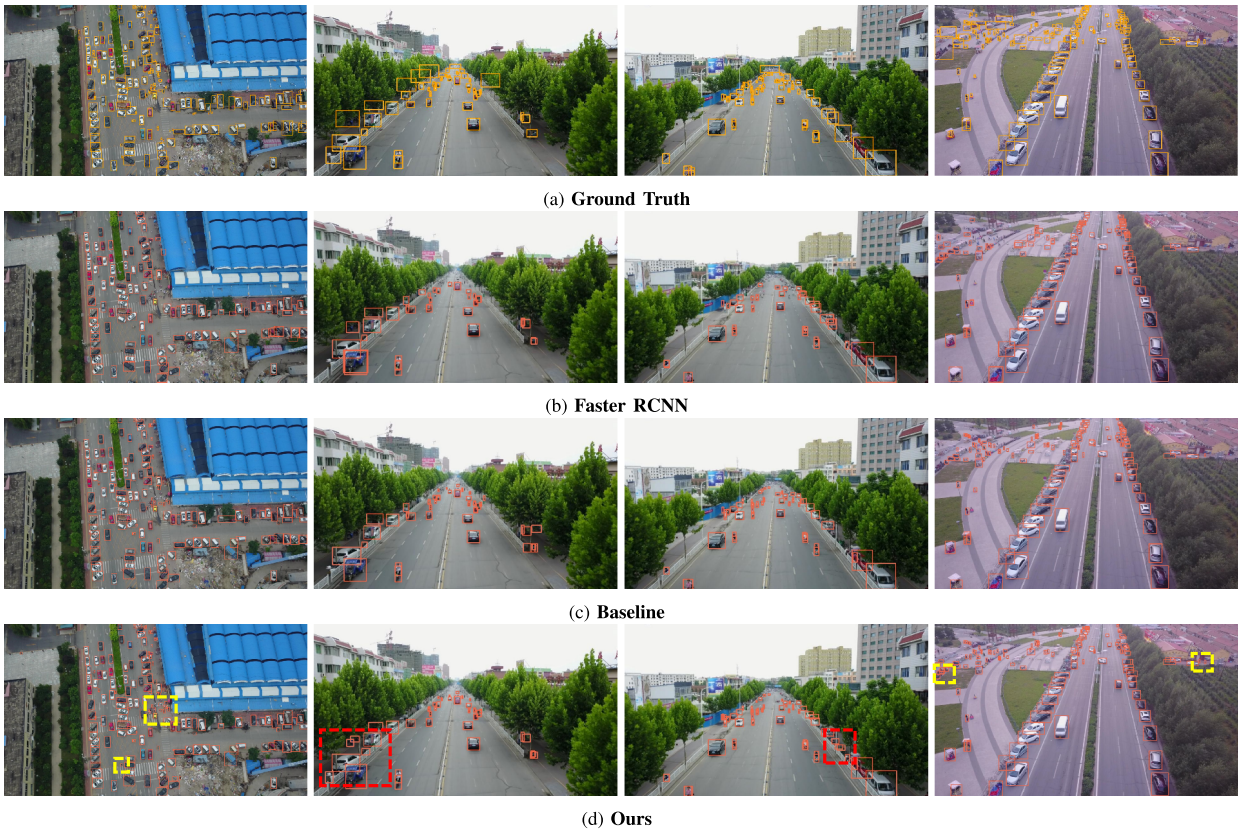

(a) **Ground Truth**

(b) **Faster RCNN**

(c) **Baseline**

(d) **Ours**

Fig. 6. Qualitative comparison results. The bright yellow and red dashed boxes show the superior performance of our method at detecting small-scale objects. (a) Ground truth. (b) Faster RCNN. (c) Baseline. (d) Ours.

TABLE III

RESULTS OF OUR PROPOSED BAFNET COMPARED WITH THOSE OF SOTA APPROACHES ON THE DIOR DATASET. RED AND BLUE: TOP TWO PERFORMANCES

Method	Backbone	APL	APO	BF	BC	BR	CH	DAM	ESA	ETS	GC	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
FCOS [25]	ResNet-101	63.2	81.9	75.3	85.9	40.7	77.7	63.6	85.1	65.0	83.8	74.7	54.4	57.4	70.3	65.1	59.0	83.8	56.2	37.9	83.1	68.2
RetinaNet [20]	ResNet-101	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
LO-Det [54]	MobileBetV2	72.6	65.0	76.7	84.7	33.7	73.7	56.8	75.9	57.5	66.3	68.0	60.9	51.5	88.6	68.0	64.3	86.3	47.6	42.4	76.7	65.9
YOLOv3 [42]	DarkNet53	60.9	55.6	65.1	82.8	22.5	68.5	40.2	58.6	55.2	61.8	61.6	50.6	44.9	82.2	54.1	56.7	80.1	34.8	32.9	70.1	57.0
SSD [21]	VGG-16	59.5	72.7	72.4	75.7	29.7	65.8	56.6	63.5	53.1	65.3	68.6	49.4	48.1	59.2	61.0	46.6	76.3	55.1	27.4	65.7	58.6
RSADet [55]	DLA-34	73.6	86.0	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5	72.2
AFDet [56]	DLA-34	82.4	81.5	81.9	89.8	51.7	74.9	58.7	84.2	73.3	79.5	81.0	44.2	62.0	77.8	63.2	76.9	91.0	62.5	59.3	87.1	73.2
CANet [57]	ResNet-101	70.3	82.4	72.0	87.8	55.7	79.9	67.7	83.5	77.2	77.3	83.6	56.0	63.6	81.0	79.8	70.8	88.2	67.6	51.2	89.6	74.3
CenterNet [24]	DLA-34	67.8	68.8	75.0	83.2	44.6	67.6	44.7	71.0	67.7	66.6	72.5	41.6	56.3	72.0	54.2	70.3	85.3	48.6	51.6	79.2	64.4
CSFF [58]	ResNet-101	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
FSoD-Net [59]	VGG-16	88.9	66.9	86.8	90.2	45.5	79.6	48.2	86.9	75.5	67.0	77.3	53.6	59.7	78.3	69.9	75.0	91.4	52.3	52.0	90.6	71.8
SDPNet [60]	CSPDarknet53	86.8	81.4	83.4	90.5	52.7	78.1	63.2	87.8	73.6	81.9	85.6	65.2	64.7	80.4	82.6	77.8	87.5	56.1	59.9	84.7	76.2
Ours	ResNet-50	66.9	91.6	78.3	89.5	53.5	81.4	77.7	89.4	80.9	83.8	85.8	63.3	65.9	89.9	79.9	75.4	88.5	72.7	56.4	91.3	78.1

3) Experiment on DIOR: Table III provides a detailed comparison of the performance of BAFNet with that of other methods on the DIOR dataset. An analysis of the table clearly reveals that BAFNet achieves the highest scores in multiple categories. Particularly in dealing with challenging categories such as expressway-toll-stations (ETSs) and windmills (WMs), where features are relatively subtle, BAFNet outperforms the other methods in terms of performance. Additionally, BAFNet maintained competitive performance in other categories. Compared with other SOTA detectors, our method achieves a combined performance mAP of 78.1%, which is 1.9% greater than that of the second-ranking SDPNet [60]. These results indicate that BAFNet demonstrates robustness and generalization capabilities when handling complex backgrounds and diverse object detection tasks.

TABLE IV

RESULTS OF OUR PROPOSED BAFNET COMPARED WITH THOSE OF SOTA APPROACHES ON THE LEVIR-SHIP DATASET

Method	AP_{50}	Method	AP_{50}
Efficient-D0 [3]	71.3	Efficient-D2 [3]	80.9
RetinaNet [20]	74.9	FCOS [25]	75.5
SSD [21]	52.9	CenterNet [24]	77.7
Faster R-CNN [18]	70.8	DRENet [61]	82.4
ORFENet [62]	83.3	**ours**	**84.2**

4) Experiment on LEVIR-Ship: As shown in Table IV, our proposed BAFNet achieves an AP_{50} of 84.2% on the LEVIR-Ship dataset, surpassing other SOTA methods. In particular,

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TABLE V

ABLATION EXPERIMENTS ON THE VISDRONE VALIDATION SET. DSAM INDICATES THE DUAL-STREAM ATTENTION MODULE. BAB INDICATES THE BOUNDARY-AWARE BRANCH. THE BOLD FONT INDICATES THE BEST RESULT

DSAM	BAB	PT	PE	BC	CA	VA	TK	TC	AT	BU	MO	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
		23.4	15.1	12.8	56.3	35.3	29.5	19.2	11.0	42.5	24.7	27.0	45.1	27.5	17.7	39.2	44.5
✓		29.8	18.9	14.4	**59.4**	35.4	27.9	21.1	12.6	46.0	28.5	29.3	51.2	29.1	21.0	39.6	44.6
	✓	29.8	**19.7**	**14.9**	**59.4**	36.6	29.8	21.2	**12.7**	44.8	28.7	29.7	51.6	29.4	21.2	39.8	**46.6**
✓	✓	**30.3**	19.6	14.8	59.3	**37.0**	**30.1**	**22.6**	12.2	**48.7**	**28.8**	**30.8**	**52.6**	**30.9**	**22.4**	**41.0**	43.2

compared with methods such as Efficient-D0 [3], Efficient-D2 [3], RetinaNet [20], FCOS [25], CenterNet [24], and Faster R-CNN [18], BAFNet has a more pronounced performance advantage. This proves that BAFNet is better at capturing objects in various complex scenarios, thereby achieving higher detection performance.

C. Ablation Study

This section delves into ablation studies aimed at assessing the impact of individual components—the DSAM and boundary-aware auxiliary branch (BAB)—on our approach. Additionally, we explored and determined the optimal configuration for our approach. All the experiments in this section are conducted on the VisDrone validation set.

1) Effectiveness of the Dual-Stream Attention Module: We first evaluated the effectiveness of the proposed DSAM, which captures more discriminative features of small objects. As shown in Table V, by introducing the DSAM into the baseline, the detection AP improves by 2.3% (from 27.0% to 29.3%). AP_{50} increases by 6.1% (from 45.1% to 51.2%), and the APs improve by 3.3% (from 17.7% to 21.0%). The performance improvement is most significant for small objects. This finding demonstrates that the loss of key detail information about small objects during feature extraction and fusion can degrade the detection performance. Our DSAM leverages the highest-level semantic features to capture more meaningful foreground and background cues, which are then fused with low-level features rich in spatial details to enhance object representation.

We conducted further investigations to explore different application strategies for the DSAM. In our method, we leveraged high-level feature layers to generate FPAM and BPAM, as these layers encapsulate richer semantic information through multilayer convolution and pooling operations. This semantic information is crucial for distinguishing the foreground from the background and recognizing target categories. Furthermore, considering that small objects in remote sensing images can be minute and risk being lost in the highest-level feature layer P_4, we select the P_3 and P_4 feature layers in our ablation study to determine the optimal strategy for generating FPAMs. As shown in Table VI, when the P_4 feature layer is used, our AP score peaks at 30.8%, with a small object detection accuracy of 22.4%. This indicates that the highest-level feature layer, P_4, provides highly discriminative feature representations for capturing foreground and background cues. It effectively captures fine-grained details, which are more conducive to small object detection tasks. Therefore, we choose to utilize the P_4 feature layer for generating FPAM and BPAM.

TABLE VI

COMPARISON OF DIFFERENT FEATURE LAYERS OF DSAM ON VISDRONE

P_3	P_4	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
		27.0	45.1	27.5	17.7	39.2	44.5
✓		29.7	51.4	29.2	21.3	40.1	**46.0**
	✓	**30.8**	**52.6**	**30.9**	**22.4**	**41.0**	43.2

TABLE VII

COMPARISON OF DIFFERENT BOUNDARY EXTRACTION METHODS FOR THE BOUNDARY-AWARE BRANCH ON VISDRONE. BAB INDICATES THE BOUNDARY AGGREGATION MODULE

BAM	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Canny	29.8	52.1	29.6	21.7	39.7	41.1
Laplacian	**30.8**	**52.6**	**30.9**	**22.4**	**41.0**	**43.2**

2) Effectiveness of the Boundary-Aware Branch: After introducing the boundary-aware branch into the baseline model, the AP_{50} and AP on the VisDrone dataset improved by 6.5% and 2.7%, respectively, as shown in Table V (Row 3). On this basis, the joint use of the DSAM module and the boundary-aware auxiliary branch outperformed the use of either method alone in terms of small object detection. As presented in Table V (Row 4), the overall AP metric increased by 3.8%, AP_{50} by 7.5%, and notably, AP_s (for small objects) by 4.7%, demonstrating the effectiveness of the boundary-aware branch in enhancing the detection accuracy of small objects. By leveraging boundary information, the model can retain more discriminative object features, which facilitates more precise localization and identification of objects.

3) Effectiveness of the Boundary Aggregation Module: In the boundary-aware branch, the boundary aggregation module is first utilized to extract boundaries. To validate the effectiveness of the Laplacian pyramid, we compare it with the commonly used edge detection method, Canny, as shown in Table VII. The experimental results indicate that the Laplacian pyramid method achieves better results across all the metrics. We attribute this to its ability to extract multiscale boundary information, which helps capture finer-grained details, thereby enabling more precise detection.

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TABLE VIII
ABLATION EXPERIMENTS WITH HYPERPARAMETER α ON THE VISDRONE DATASET

BAB	α	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
	-	29.3	51.2	29.1	21.0	39.6	44.6
✓	0.1	29.7	51.8	29.6	21.9	39.4	44.7
✓	0.3	30.1	52.2	29.8	21.7	40.0	**45.8**
✓	0.5	**30.8**	**52.6**	**30.9**	**22.4**	**41.0**	43.2
✓	0.8	30.2	52.4	30.2	22.1	40.4	41.2
✓	1.0	30.3	52.5	30.2	22.0	40.9	44.5

4) Analysis of Hyperparameters for Boundary Loss Weights: In the training process, we employ the hyperparameter α to regulate the effect of the boundary loss. To validate the impact of α on model performance, we conducted experiments with various α settings. The detection results on the VisDrone dataset are presented in Table VIII, indicating that as α increases, the model's detection performance generally tends to first increase and then decrease. This could be attributed to the fact that too small an α value results in an overly weak contribution from the boundary loss, whereas an excessively large α tends to dominate the overall loss, neglecting classification and detection losses, leading to a decrease in the AP. Consequently, we choose to set α to 0.5, which achieves the highest AP and AP_s.

V. CONCLUSION

This article proposes BAFNet for the challenging task of small object detection in remote sensing. Remote sensing images often contain distant and low-resolution small objects. The BAFNet framework incorporates several key components specifically designed to enhance the detection capability of small objects. The DSAM module improves the representation of foreground features, capturing discriminative features of small objects. The boundary-aware supervision strategy retains detailed information, which is crucial for distinguishing small objects from complex backgrounds. Evaluations conducted on the AI-TOD, VisDrone, DIOR, and LEVIR-Ship datasets demonstrate that BAFNet has significant advantages in terms of detection accuracy, outperforming existing methods and providing a powerful solution to the complex problem of small object detection in remote sensing imagery. In the future, we will continue to study the processing of noise and delve into a detection algorithm that is more sensitive to small objects.

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